

Probabilistic Artificial Intelligence

Final Project Report

Probabilistic forecasting of Portuguese football match results
and utility analysis with betting odds

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Degree project report

This report follows the structure of the project statement: Bayesian Networks, probabilistic classification, uncertainty quantification, calibration, Bayesian Optimization, utility analysis and Reject Option.

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Abstract

In this project I developed a probabilistic system to analyse football matches from the Portuguese league. The goal was not only to predict whether the home team wins, the match ends in a draw, or the away team wins, but also to study the uncertainty of those predictions and compare the model probabilities with market odds.

The workflow followed the project statement. First, I prepared and explored the dataset. Then I built and analysed Bayesian Networks, trained a Naive Bayes probabilistic classifier, applied Conformal Prediction, evaluated probability calibration, used Bayesian Optimization, and finally studied utility and Reject Option using betting odds. The main conclusion is that the model gives reasonable results for a difficult problem like football, but its probabilities must be used carefully, especially when they are used for betting or decision making.

1. Introduction and project objective

The objective of the project is to build a probabilistic forecasting system for football match results. Each match has three possible outcomes: home win (H), draw (D), or away win (A). Unlike a simple classifier that only returns one label, this project focuses on probabilities. For example, saying “home win with probability 0.70” is much more informative than only saying “home win”.

The project is based on three main ideas. First, Bayesian Networks are used to represent relationships between variables such as recent form, attack level, head-to-head dominance and the favourite according to the odds. Second, a probabilistic classifier, Naive Bayes, is trained and evaluated not only with accuracy but also with uncertainty and calibration. Third, the predicted probabilities are compared with market odds in order to study utility and decision making under uncertainty.

During the project I tried to keep a balance between performance and interpretability. Football is a noisy domain: a good model can still fail because many events are random or are not present in the dataset. For that reason, the most important objective was not to create a perfect predictor, but to build a complete and coherent probabilistic workflow and to interpret its limitations.

2. Dataset and data preparation

The dataset contains 5,583 matches from the Portuguese football league. Each row corresponds to one match and includes information about teams, goals, odds, recent form, head-to-head history and discretised variables. The target variable is result, with three possible values: H, D and A.

Result	Number of matches	Percentage	Interpretation
H	2,466	44.17%	Home win is the most frequent result.
A	1,712	30.66%	Away wins are less frequent than home wins.
D	1,405	25.17%	Draw is the least frequent class.

This distribution is important because the data is not perfectly balanced. The model sees more examples of home wins than draws. Therefore, accuracy alone is not enough: a model could look acceptable because it predicts frequent classes, while still performing badly on draws.

I also checked missing values in all columns. The final dataset had no missing values, so it was possible to continue without removing matches or adding extra imputation steps.

2.1. Variables used

For the Bayesian Network I used the discretised variables, because Bayesian Networks work naturally with categorical variables and conditional probability tables.

Variable	Meaning
ht_form / at_form	Recent form of the home and away teams.
ht_attack / at_attack	Recent attacking level of the home and away teams.
joint_attack	Combined attacking profile of both teams.
h2h_dominance	Historical dominance in direct confrontations.
odds_favorite	Favourite according to the market odds.
result	Final result of the match: H, D or A.

For the classification part I used variables related to odds, recent form, goals, wins and head-to-head information. This separation was useful because the Bayesian Network was mainly used for probabilistic reasoning, while the classifier was used for prediction and uncertainty analysis.

2.2. Exploratory analysis

Variable group	Main observation	Why it matters
Recent form	Most teams are in intermediate categories such as below average or average.	Extreme states such as excellent are less frequent, so some CPT rows have fewer examples.
Attack level	The average attack category is dominant for both home and away teams.	The model has more evidence for average situations than for strong attacks.
Head-to-head dominance	Most matches are classified as balanced.	In many pairings there is no clear historical dominance.
Odds favourite	The home team is the favourite in most matches.	This matches the fact that home wins are also the most frequent result.

This first analysis was important because it helped me understand the data before training the models. In particular, the existence of rare combinations explains why Bayesian parameter estimation can be more stable than pure frequency estimation.

3. Part 1: Bayesian Networks

The first part of the project was to build a Bayesian Network for the football domain. A Bayesian Network allows us to represent a large joint probability distribution as a product of smaller local probability tables. In practical terms, this lets us model dependencies between variables without estimating one huge table with all possible combinations.

3.1. Manual Bayesian Network

The first network was built using Knowledge Engineering, meaning that the structure was chosen before learning the parameters. The manual network has 8 nodes and 10 directed edges.

Edge	Reason for including it
ht_attack -> ht_form	If the home team has been scoring more goals, it can help explain better recent form.
at_attack -> at_form	The same idea applies to the away team.
ht_attack -> joint_attack	joint_attack is built using the home attack level.
at_attack -> joint_attack	joint_attack is also built using the away attack level.
ht_form -> odds_favorite	The market can reflect the recent form of the home team.
at_form -> odds_favorite	The market can also reflect the recent form of the away team.

h2h_dominance -> odds_favorite	Historical dominance may influence how the market sees the match.
ht_form -> result	The home team's recent form can affect the final result.
at_form -> result	The away team's recent form can also affect the final result.
odds_favorite -> result	The market favourite is usually related to the final outcome.

One important decision was not to connect ht_attack, at_attack and joint_attack all directly to result. Since joint_attack already contains information from both attacks, adding all three as direct parents of result would be redundant. Also, scoring many goals recently does not guarantee winning the next match. For that reason, attack was mostly allowed to influence the result indirectly through recent form.

3.2. Automatic structure learning

After the manual network, I applied automatic structure learning using Hill Climbing Search. I tested two scoring functions: BIC and BDeu. I also limited the maximum number of parents per node to 3, because the goal was not only to fit data but also to keep the network understandable.

Structure	Nodes	Edges	Main interpretation
Manual	8	10	Most interpretable, because the edges follow football reasoning.
Hill Climbing + BIC	8	10	Keeps some intuitive edges, but also adds less natural statistical dependencies.
Hill Climbing + BDeu	8	9	Slightly simpler, but visually and conceptually less convincing for the football domain.

The BIC structure shared several edges with the manual network, including at_attack -> at_form, ht_attack -> ht_form, h2h_dominance -> odds_favorite and odds_favorite -> result. This means the algorithm found some relationships that agree with the initial intuition.

However, the learned structure also produced edges such as joint_attack -> ht_attack and odds_favorite -> joint_attack. These should not be read as football causality. They are statistical dependencies that help the algorithm factorise the data. For the rest of the work I kept the manual network as the main model, and used the learned network as a comparison.

3.3. D-separation

D-separation was used to understand which conditional independencies are implied by the manual network. This step is useful because a Bayesian Network is not only a prediction tool; it also makes assumptions about which variables stop giving extra information when other variables are known.

Case	Result	Interpretation
ht_attack and at_attack with no evidence	d-separated	The recent attack of the home team and the away team are independent before observing other variables.
ht_attack and odds_favorite given ht_form	d-separated	Once home form is known, home attack does not add direct extra information to odds_favorite through this path.
ht_form and at_form given odds_favorite	d-connected	This is a V-structure: observing the market favourite opens a connection between both forms.

The third case is the most interesting one. The structure $ht_form \rightarrow odds_favorite \leftarrow at_form$ is a collider. If $odds_favorite$ is not observed, the two forms are not connected through this path. Once the favourite is observed, knowing the home form can give indirect information about the away form, because both help explain why the market selected a favourite.

3.4. Parameter learning: MLE and Bayesian estimation

After fixing the manual structure, I learned the conditional probability tables. I used two methods: Maximum Likelihood Estimation (MLE) and Bayesian estimation with a BDeu prior.

MLE estimates probabilities directly from observed frequencies. This is simple and useful, but it can create probabilities equal to 0 or 1 when some parent configuration appears only a few times. In football, this is risky: a result should not be considered impossible just because it did not appear in a very small group of examples.

Comparison point	MLE	Bayesian estimation with BDeu
Frequent configurations	Very similar to Bayesian estimation.	Very similar because the data dominates the prior.
Rare configurations	Can produce extreme probabilities.	Produces smoother and more cautious probabilities.
Extreme rows in result CPD	8 extreme rows.	0 extreme rows.
Chosen for final inference	Used as reference.	Used as the preferred model for queries.

For example, in one rare parent configuration with only two examples, MLE produced $P(A)=0$, $P(D)=0.5$ and $P(H)=0.5$. Bayesian estimation changed this to $P(A)=0.0108$, $P(D)=0.4946$ and $P(H)=0.4946$. The second result is more cautious because it avoids saying that an away win is impossible.

3.5. Probabilistic inference

Finally, I used the trained Bayesian Network to answer probabilistic queries. I used the Bayesian-estimated network because it is more stable when there are rare combinations of evidence.

Query	Evidence	Probabilities	Interpretation
Q1	Home team in good form and favourite in the odds	$P(A)=0.1164$, $P(D)=0.1884$, $P(H)=0.6952$	The network strongly favours a home win, which is coherent with the evidence.
Q2	Both teams with average form	$P(A)=0.3404$, $P(D)=0.2706$, $P(H)=0.3890$	The match is more balanced, with a small advantage for the home team.
Q3	Home team has head-to-head dominance	$P(A)=0.2051$, $P(D)=0.2485$, $P(H)=0.5464$	The most likely result is again a home win.
Critical case	A contradictory or rare evidence configuration	$P(A)=0.4946$, $P(D)=0.0108$, $P(H)=0.4946$	This is not very realistic from football intuition because the draw becomes too unlikely.

Overall, the Bayesian Network is useful for reasoning with evidence. In many cases the answers are coherent, but some rare situations produce probabilities that should be interpreted critically.

4. Part 2: Probabilistic classification and uncertainty

The second part of the project focused on probabilistic classification. The main model was Naive Bayes, because it was explicitly required by the project and because it naturally returns class probabilities.

4.1. Temporal split

I split the data by date instead of using a random split. This is more realistic for football, because in a real setting we train on past matches and evaluate on future matches.

Split	Size	Period	Purpose
Train	3,628	2006-01-06 to 2019-09-22	Train Naive Bayes.
Calibration	950	2019-09-22 to 2022-10-22	Calibrate probabilities and Conformal Prediction.
Test	1,005	2022-10-22 to 2026-01-26	Final evaluation.

The calibration block was placed just before the test block. The idea is that recent calibration errors are more representative of the test period than very old errors.

4.2. Naive Bayes results

Metric	Value
Accuracy	0.5483
Balanced accuracy	0.4998
Log Loss	1.1452

Class	Precision	Recall	F1-score	Support
A	0.57	0.59	0.58	323
D	0.33	0.22	0.26	236
H	0.60	0.70	0.65	446

The model correctly predicts around 55% of the matches. For football this is reasonable, but it must be interpreted carefully. Balanced accuracy is lower, which means the model does not perform equally well on all classes.

The main problem is the draw class. Out of 236 real draws, only 51 were predicted as draws. This is expected to some extent because draws are the least frequent class and are also difficult to separate from close home or away wins.

True \ Predicted	A	D	H
A	190	44	89
D	70	51	115
H	75	61	310

4.3. Conformal Prediction

I then applied Split Conformal Prediction to the Naive Bayes model. Instead of always returning one class, the model returns a set of possible classes. For example, it can return {H}, {H, D}, or {A, D, H}. A larger set means more uncertainty.

Metric	Value	Meaning
Target coverage	0.9000	Desired probability that the true label is inside the prediction set.
Empirical coverage	0.9025	The achieved coverage is very close to the target.
Mean prediction set size	2.2637	The average set contains more than two classes.
Set size distribution	257 of size 1; 226 of size 2; 522 of size 3	More than half of the matches contain all three classes.

The method achieves the intended coverage, which is positive. However, many prediction sets contain all three outcomes, so the result is safe but sometimes not very informative. This is still useful because the set size gives a practical measure of uncertainty, and it helps justify the Reject Option later.

4.4. Probability calibration

Since the probabilities are later used for odds, utility and Reject Option, it is not enough to evaluate only the predicted class. I used Reliability Diagrams, Brier Score, Log Loss and ECE. Then I applied Platt Scaling as a post-hoc calibration method.

Model	Brier Score	Log Loss	ECE	Interpretation
Naive Bayes original	0.6135	1.1452	0.1241	Probabilities are not very well calibrated.
Naive Bayes + Platt Scaling	0.5643	0.9556	0.0340	Probabilities improve clearly after calibration.

Calibration clearly improved the probabilistic quality of the model. Log Loss decreased from 1.1452 to 0.9556, and ECE decreased from 0.1241 to 0.0340. This means that calibrated probabilities are closer to the real observed frequencies. For the utility part, I therefore used the calibrated probabilities.

Even after calibration, the draw class remained the most difficult one. This is an important limitation because football draws are naturally ambiguous.

4.5. Bayesian Optimization

Bayesian Optimization was applied to tune the Naive Bayes hyperparameters. I optimized alpha and fit_prior, using negative Log Loss as the objective. I chose Log Loss because the goal was to improve probability quality, not only the final class label.

Model	Accuracy	Balanced accuracy	Log Loss
Baseline Naive Bayes	0.5483	0.4998	1.1452
Optimized Naive Bayes	0.5502	0.4965	1.1085

The best configuration was alpha = 10.0 and fit_prior = True. The improvement was small but coherent: accuracy increased slightly and Log Loss improved. Balanced accuracy decreased a little, so the optimization did not improve all metrics at the same time. My interpretation is that Bayesian Optimization was useful, but it did not radically change the behaviour of Naive Bayes because the model is simple and the problem is complex.

5. Part 3: Utility, odds and decision making

The third part changes the main question. Now the question is not only “does the model predict correctly?”, but also “can its probabilities be useful when compared with the odds?” For that reason I separated predictive evaluation from utility evaluation.

5.1. Comparing model probabilities with market odds

First, I converted the odds into implied probabilities. Since the three implied probabilities can sum to more than 1 because of the bookmaker margin, I normalised them.

$$\text{raw_H} = 1 / \text{h_odds}, \text{raw_D} = 1 / \text{d_odds}, \text{raw_A} = 1 / \text{a_odds}$$

$$\text{P_market(H)} = \text{raw_H} / (\text{raw_H} + \text{raw_D} + \text{raw_A})$$

Metric	Value	Interpretation
Test matches	1,005	Matches used for utility analysis.

Agreement cases	837	Model and market choose the same favourite.
Disagreement cases	168	Model and market choose different favourites.
Mean model confidence	0.5375	Average confidence of the model.
Mean market confidence	0.5387	Average confidence of the market after normalisation.
Mean maximum disagreement	0.1021	Average largest probability gap between model and market.

The model and the market agree in most matches. This suggests that the model captures a signal similar to the one present in the odds, especially when there is a clear favourite. The interesting cases are the disagreements, because they show where the model sees the match differently from the market.

5.2. Expected utility and value bets

For each possible bet I calculated the expected value. If we bet one unit on outcome k , I used:

$$EV(k) = P_{\text{model}}(k) \times \text{odds}(k) - 1$$

If $EV(k)$ is positive, the model thinks that the outcome is more likely than what the odds suggest. Initially I used $EV > 0.05$ as the threshold for a value bet.

Threshold	Value bets found	Interpretation
$EV > 0.05$	813 of 1,005 matches (80.90%)	Too many cases; not realistic for a real betting strategy.
$EV > 0.15$	Used in the final hybrid strategy	More demanding and therefore more cautious.

The very high number of value bets with $EV > 0.05$ is a warning sign. Mathematically the condition is satisfied, but in a real betting market it is not realistic to expect value in more than 80% of matches. This probably means that the model probabilities, even after calibration, are still not strong enough to be used directly as betting probabilities.

5.3. Reject Option and deferring to odds

Reject Option means that the system is not forced to decide in every match. If the model is uncertain, it can abstain or use another source. In this project the external source is the betting market.

Final decision rule:

1. If model confidence is at least 0.55, use the model prediction.
2. If the model is not confident but there is a strong value bet with $EV > 0.15$, use the value bet decision.
3. If there is no strong value bet but the market confidence is at least 0.55, defer to the odds.
4. If none of the previous conditions is satisfied, reject or abstain.

Decision type	Number of matches
Use model	507
Value bet	228
Defer to odds	3
Reject	267

Most accepted decisions come from the model itself. Direct deferral to odds appears only 3 times because many cases are already accepted earlier by model confidence or by strong value bet.

5.4. Strategy comparison

Strategy	Coverage	Accuracy on accepted cases	Interpretation
Always use model	1.0000	0.5502	Predicts every match but has moderate accuracy.
Reject Option - model confidence	0.5045	0.6824	Much more accurate, but only predicts about half of the matches.
Hybrid: model + value bet + odds + reject	0.7343	0.5556	Predicts more matches, but accuracy is close to always using the model.

The best strategy for improving the accuracy of accepted predictions is the pure confidence-based Reject Option. It raises accuracy from 0.5502 to 0.6824, but it only decides on about half of the matches. The hybrid strategy increases coverage to 0.7343, but its accuracy drops close to the always-use-model baseline.

The practical conclusion is that uncertainty is useful for deciding when to abstain. However, the value bet component should be treated as exploratory analysis, not as a reliable real betting strategy.

6. Critical analysis

The project shows that probabilistic models are useful because they give more information than a single predicted class. They allow us to study uncertainty, calibration and utility. However, they also have important limitations.

Strength	Explanation
Complete probabilistic workflow	The project covers Bayesian Networks, classification, uncertainty, calibration and utility.
Interpretability	The manual Bayesian Network is easy to explain using football reasoning.
Uncertainty-aware decisions	Reject Option improves performance when the model is confident.
Calibration step	Platt Scaling improves the reliability of predicted probabilities.

Limitation	Explanation
Football is noisy	Many factors such as injuries, line-ups or red cards are not in the dataset.
Draws are hard	The model performs clearly worse on the draw class.
Rare Bayesian Network configurations	Some CPT rows have few examples and can lead to unstable probabilities.
Value bets are not fully reliable	The model finds too many opportunities, which suggests probabilities are still imperfect for betting.

The main lesson is that good accuracy is not enough in a probabilistic project. If probabilities are used for decisions, it is necessary to check calibration, uncertainty and utility. In this project, the model is useful as an analytical tool, but the final decisions must remain cautious.

7. Conclusions

In this project I developed a complete Probabilistic Artificial Intelligence workflow applied to football match results. First, I prepared the dataset and analysed the variables. Then I built a manual Bayesian Network,

compared it with automatically learned structures, analysed D-separation, and learned parameters using MLE and Bayesian estimation.

The manual Bayesian Network was more interpretable than the automatically learned structures. The learned networks detected some reasonable relationships, but also produced edges that were difficult to justify from a football point of view. For that reason, the manual network was the best option for explaining the domain.

In the classification part, Naive Bayes achieved accuracy close to 55%. This is reasonable for football, but the performance was not equal across classes, especially for draws. Conformal Prediction produced coverage close to 90%, but many prediction sets were large, showing that the model is often uncertain.

Calibration was one of the most important steps. Platt Scaling improved Brier Score, Log Loss and ECE, so the calibrated probabilities were more appropriate for comparison with odds. Bayesian Optimization also improved Log Loss slightly, although it did not radically change the model.

In the utility part, comparing the model with market odds helped detect disagreements. However, value bets were not reliable enough as the main rule. The strongest decision strategy was the confidence-based Reject Option: when the model only decides in cases where it is more confident, the accuracy of accepted predictions increases clearly.

My final conclusion is that the system is useful as a probabilistic analysis and decision-support tool, but it should not be interpreted as a real betting strategy. The most important result of the project is learning how to combine prediction, uncertainty, calibration and utility in a critical way.

8. Deliverables

Together with this report, the project deliverables include the code and the processed data used in the different phases. The notebook contains data loading, Bayesian Network construction, structure and parameter learning, probabilistic inference, Naive Bayes training, calibration, Conformal Prediction, Bayesian Optimization and the utility analysis with Reject Option.